

J.A.R.V.I.S. Academy · Session 3 — Memory · Expert session

The two-hour route — persistence first, then meaning

0:00–0:05 · Opening

Scope, the standard of performance, and the one-line thesis of the session.

0:05–1:00 · Act I

Memory persistence — slides 3–24. Three classes, the storage map, the durability boundary, the recall cycle.

1:00–1:50 · Act II

Memory semantics — slides 25–44. Curation, reconciliation, continuity, and the publication boundary.

1:50–2:00 · Closing

Slides 45–48. The two acts in one line each, the guarantees, and the cold self-check.

- Classify any fact into its persistence class, cold
- Reconcile a correction without appending a contradiction
- State the storage map and each read policy
- Defend the publication boundary and its three nevers

HgX · Thursday 2026-10-22 · 6:00–8:00 AM America/Toronto · Sources: vision white paper v1.2, section 05; technical white paper, section 8.

Act I — memory persistence: where state lives

Act I · Memory persistence · 0:05–1:00 · Slides 3–24

> **“Memory preserves identity; live systems preserve truth.”**

The act’s question

Where does each kind of state live, how long does it live there, and under what policy is it read back? Persistence is architecture, not storage trivia.

The act’s answer, in advance

Three persistence classes, four storage addresses, one durability boundary, and a four-move cycle that runs before any work on prior state.

Three persistence classes govern every kind of state

Class	Contents	Read policy
Durable memory	Stable facts, preferences, commitments, named relationships.	Load when prior decisions or history matter.
Working state	Run logs, ledgers, watermarks, drafts, source snapshots.	Read from the owning goal or service.
Live source	Prices, schedules, account state, current records.	Re-read at decision time; do not trust memory.

The technical paper's three persistence classes, verbatim. Every kind of state the estate holds belongs to exactly one class — and the class, not convenience, sets the read policy.

Durable memory holds what changes by decision, not by time

Durable memory

Class 1 of 3 · The identity store

Contents — verbatim

Stable facts, preferences, commitments, named relationships.

Read policy — verbatim

Load when prior decisions or history matter.

Durable memory changes slowly and by decision, not by the passage of time. It is loaded at recall — before acting on prior work — because it carries how the estate works.

Working state belongs to the work that produced it

Working state

Class 2 of 3 · The run's own record

Contents — verbatim

Run logs, ledgers, watermarks, drafts, source snapshots.

Read policy — verbatim

Read from the owning goal or service.

Working state belongs to the work, not to the estate's identity. It is read from its owner — the goal or service that produced it — never promoted into durable memory by accident of storage.

Live source is re-read at decision time — memory is not trusted

Live source

Class 3 of 3 · Never an authority from memory

Contents — verbatim

Prices, schedules, account state, current records.

Read policy — verbatim

Re-read at decision time; do not trust memory.

Live-source facts are not stored as authority at all. Memory may hold where the source can be found — it may never hold the value as settled. The read happens at decision time, or the decision is made on a memory of the world, not the world.

Checkpoint — classify six facts, cold

1	HgX's communication style — which class?	2	This morning's account balance — which class?
3	A run log's watermark — which class?	4	A standing approval rule — which class?
5	A draft awaiting review — which class?	6	Today's schedule — which class?

Classify cold. Name the class and the read policy that follows from it — the policy is the answer, the label is only its name.

The storage map — four addresses hold the estate’s memory

~/MEMORY.md	The curated long-term record — one file, deliberately small, deliberately maintained.
~/memory/	Dated notes plus the people and group indexes, with a detailed page per named subject.
bank/ · index/	Runtime-owned memory infrastructure — the machinery under the record, not the record itself.
Side chats	Independent context scopes, one per bot duty — each role keeps its own memory scope.

The technical paper's storage map. Four addresses hold everything the estate remembers; the next four slides take one address each.

~/MEMORY.md — the curated long-term record

~/MEMORY.md

Storage map · Address 1 of 4

“Curated long-term record: facts, preferences, commitments, and durable operating decisions.”

This is durable memory's home: the file loaded when prior decisions or history matter. Its discipline is curation — entries earn their place, and a correction reconciles the entry rather than adding a rival (Act II, slide 27).

~/memory/ — dated notes and the relationship indexes

~/memory/

Storage map · Address 2 of 4

> **“Dated notes plus people and group indexes. Detailed relationship pages consolidate facts for the named subject.”**

Where ~/MEMORY.md holds the settled record, ~/memory/ holds its supporting detail: the dated trail, and one consolidated page per person and per group — consolidation, so a subject's facts live in one place and are corrected in one place.

bank/ · index/ — infrastructure the agent may read, never own

bank/ · index/

Storage map · Address 3 of 4

> **“Runtime-owned memory infrastructure. Read-only to the operating agent.”**

The machinery under the record belongs to the runtime, not to the agent. Read-only is a boundary, not an inconvenience: the agent works through the memory it is given and never rewrites the infrastructure that serves it.

Side chats — independent scopes, aligned to duties

Side chats

Storage map · Address 4 of 4

> **“Independent context scopes aligned to bot duties. A bot’s work and memory do not cross into another bot’s role.”**

Scope follows duty

Each bot’s memory scope is drawn around its job. Relevance rises because the scope excludes everything the role does not need.

Discretion follows scope

A narrow scope is also a disclosure control: a role cannot repeat what its memory never held.

The limit, stated by the paper itself: procedural, not cryptographic

> **“Security limitation. Side-chat isolation is procedural. Separate roles, conversations, and memory scopes improve relevance and discretion, but do not establish cryptographic isolation on a shared machine.”**

The technical paper states its own limit, in its own words. Scoping is real engineering — and it is not encryption. The next slide separates what it gives from what it does not.

What procedural isolation gives — and what it does not

Procedural isolation gives

Relevance — each role works inside the context its duty needs.
Discretion — a role cannot disclose what its scope never held.
Accountability — work and memory stay attributable to the role that produced them.

It does not give

Cryptographic isolation on a shared machine. That requires independent security controls: least privilege, authenticated destinations, auditable outcomes, and credentials held only in approved secure channels.

> **“Shared infrastructure is never assumed to provide isolation merely because roles have different names.”**

~ is the persistence root — the durability boundary

Survives

State under ~ — the home directory — persists across restarts and replacements. The memory files, the workspace, the ledgers: everything the estate is accountable for lives here.

Disposable

/tmp and system paths are ephemeral by design. Anything written there is written on the assumption that it may already be gone.

The rule

Durable state is written under ~, or it is not treated as written at all. A result that matters is promoted into the persistence root before it is reported.



Act I checkpoint — persistence, defended cold

1	The three classes, with contents, from memory	2	The read policy that follows from each class
3	The storage map — all four addresses	4	Side-chat scoping: what it gives, in one line
5	The security limitation, verbatim	6	The durability boundary: ~ against /tmp

Cold, no notes. Anything missed here returns at slide 44 — and is named there.

Keep in durable memory — the vision paper's list, verbatim

- Role definitions and ownership
- HgX's preferences and communication style
- Standing principles and approval rules
- Proven methods, templates, and acceptance tests
- Decisions, commitments, and outcome history
- Where authoritative systems can be found

Re-read from live sources — the vision paper's list, verbatim

- Calendars, deadlines, and current schedules
- Balances, prices, odds, scores, and inventories
- Account, order, delivery, and document status
- Health measurements and recent trends
- Messages, approvals, and permissions
- Any fact that can change between runs

Recall, refresh, reconcile, record — the cycle before prior work

Recall

Refresh

Reconcile

Record

The vision paper's four moves, run in order, whenever work resumes on prior state. Recall loads what memory holds; refresh replaces it with what is true; reconcile names what expired; record keeps only what is durable. The next four slides take one move each, in the paper's words.

Recall — load how we work, before asking what is true

Recall

Cycle · Move 1 of 4

> **“Load the role, preferences, prior decisions, and open commitments.”**

Recall is durable memory doing its one job: restoring how we work before any question of what is true is asked. Note what it loads — role, preferences, decisions, commitments — and what it pointedly does not: no current values of anything.

Refresh — current state comes from the authoritative source

Refresh

Cycle · Move 2 of 4

> **“Read current state from the authoritative source.”**

Refresh is where live source enters: every changing or consequential fact is read from its authority — the calendar, the account, the record itself — at the moment the work resumes, not from the memory of the last session.

Reconcile — name what changed, and what expired

Reconcile

Cycle · Move 3 of 4

> **“Identify what changed and which prior assumptions expired.”**

Reconcile compares the refreshed present against the recalled past and names the difference explicitly. An expired assumption that is not named survives — silently — as a hidden premise of the new work. Act II gives this move its full legal form: the reconciliation rule.

Record — preserve only what is durable

Record

Cycle · Move 4 of 4

> **“Preserve only durable outcomes, decisions, and next obligations.”**

The word ‘only’ does the work. Record closes the cycle by refusing almost everything the run produced: outcomes, decisions, and next obligations are preserved; the rest remains working state, owned by its goal or service, or is discarded.

Act II — memory semantics: what memory means

II

Act II · Memory semantics · 1:00–1:50 · Slides 25–44

> **“Memory is curated state, not an infinite transcript”**

The act’s question

What does memory mean? Not where it sits, but what a stored fact asserts, who may overrule it, and what of it may ever leave the estate.

The act’s answer, in advance

Memory is selected, reconciled, and bounded: one current record, two authorities that never overlap, and a publication boundary enforced inside memory itself.

Curation is selection, judgment, and maintenance

Curated state

Selection: someone decided this fact earns permanence. Judgment: its wording is maintained as the estate's current position. Maintenance: corrections reconcile it, so it stays one usable record. Curated state answers questions.

Infinite transcript

Records everything, therefore selects nothing. Every superseded version survives with equal standing. The reader must re-derive the current position on every read. A transcript records everything and therefore answers nothing.

Curation is not compression. It is the continuing act of deciding what the estate currently holds to be so.

The reconciliation rule — corrections replace contradictions

> “Corrections replace contradictions — When HgX corrects a fact, the memory system reconciles the old note rather than appending a second incompatible version. The objective is a usable current record, not an archaeology of mistakes.”

The technical paper's reconciliation rule, in full. One correction, one record, reconciled — the old note does not survive as a competing version.

Appending corrupts; reconciling keeps one usable record

Appending corrupts

Two incompatible versions now coexist with equal standing. The next reader must guess which is current — or worse, merges them. Every later correction adds another stratum. The record becomes an archaeology of mistakes: complete, and unusable.

Reconciling keeps one record

The correction is applied to the note itself. What was true in the old note survives; what was corrected is replaced. One current version, answerable at a glance — a usable current record.

The continuity rule — memory consulted, sources authoritative

> “The orchestrator consults memory before acting on prior work, but uses the original files and live systems as the authority for changing or consequential facts.”

> “Memory answers 'what matters'; the source system answers 'what is true now.'”

Two authorities, two questions, no overlap

Memory answers

“what matters” — which commitments stand, which decisions bind, what the work is for. On questions of significance, memory is the authority and the source system has no vote.

The source system answers

“what is true now” — the current value, status, and state of the world. On questions of fact, the source is the authority and memory has no vote.

Two authorities, two questions, no overlap. Every memory failure in Act II is a case of one authority answering the other’s question.

The distinction is a control, not a convenience

§05

Vision white paper v1.2 · Section 05 — Memory and continuity

> **“Memory answers 'how do we work?' Live systems answer 'what is true now?' The distinction is a control, not a convenience.”**

A convenience would let the system use whichever answer is nearer. A control forbids the substitution — the distinction exists to prevent a specific error, and it binds hardest exactly when bypassing it would be easiest.

What the control prevents — three named failures

Impersonation prevented

Remembered state presenting itself as current state — the stale value wearing the authority of a fresh read.

Stale-memory decisions prevented

Actions taken on what was true at write time, in a world that has since moved — prices, schedules, statuses, balances.

Inherited assumptions prevented

A hidden premise crossing a handoff as if it were a verified fact, and governing new work unexamined.

Remembered state must never impersonate current state

§05

Vision white paper v1.2 · Section 05 — Memory and continuity

> **“Continuity is essential, but remembered state must never impersonate current state.”**

Both halves bind. Continuity is essential — the estate does not restart from ignorance each session. And the impersonation ban is absolute: remembered state may inform a decision, but it may never present itself as the current read.

The continuity contract — what a handoff must move

> “Every handoff should move the original brief, changed files or records, evidence, unresolved gaps, and the next safe action together. A new engine or specialist should be able to resume without reconstructing the entire conversation and without inheriting a hidden assumption as fact.”



Semantics checkpoint — meaning, defended cold

1	The reconciliation rule, stated as a write protocol
2	The continuity rule — both halves, in order
3	The two authorities and their two questions
4	Control, not convenience — what the control prevents
5	The continuity contract's five moving parts
6	The impersonation ban, verbatim

Cold, no notes. The boundary slides that follow are graded against these six.

The publication boundary is enforced in memory

> “memory is where the boundary is enforced in practice. Durable methods, role definitions, and principles belong to the system and are publishable. Personal memory — preferences tied to a private life, household detail, and lived history — stays in the estate and is never part of a public release.”

The vision paper places the open-source boundary inside memory practice. The system is publishable; the life it serves is not — and memory is where that line is drawn, entry by entry.

What of memory may go public — the system side

Durable methods

How the estate works when it works well — proven methods, templates, and acceptance tests. These belong to the system.

Role definitions

What each role owns, its finish line, and its boundaries. The architecture of the work, publishable by design.

Principles

The standing rules that govern conduct and approval. Doctrine is written to be inspected — inspectability is its point.



Personal memory never leaves the estate — three nevers

> **“it is the principal’s private life in durable form, so it is never staged, never sanitized into publication, and never leaves the estate.”**

Never staged

Personal memory is not prepared for an audience — not even a friendly one.

Never sanitized into publication

Redaction does not convert private memory into public material. The category, not the wording, is what is protected.

Never leaves the estate

No release, no excerpt, no exception by convenience. The boundary is absolute.

Retention discipline — enough, and no more

> **“Retention discipline: preserve enough to maintain service and accountability; do not collect sensitive context merely because storage is available.”**

Preserve enough

Service and accountability set the floor: what the estate needs to keep working, and to answer for what it did.

Do not collect

Available storage is not a reason. Sensitive context collected without need is exposure without benefit — a liability carried forever for a use imagined once.

Memory is the clearest case of the open-source boundary

> **“Memory is also the clearest case of the open-source boundary”**

Of everything the estate holds, memory is where the public system and the private life physically meet — the same store carries publishable doctrine and the principal's lived history. That is why the boundary is sharpest here: no other surface forces the system-versus-life decision on every single entry.

The failure mode has one shape — remembered state as current state

The remembered value

Acting on a price, schedule, or balance held in memory — correct when written, false when used, and presented with a fresh read's confidence.

The superseded decision

Treating a decision as current after the world — or HgX — has replaced it, because the old note was never reconciled.

The inherited assumption

A handoff that carries a hidden premise as fact, so the next engine builds on ground nobody verified.

One shape, three costumes: remembered state, worn as current state.

Drill — a fact is corrected; work the full protocol

The event	HgX corrects a standing fact held in ~/MEMORY.md. The old entry is wrong as of now.
Step 1 — Name the class	State the fact's persistence class and the storage address it lives at.
Step 2 — State the read policy	Say how the corrected fact will be read in future, and what must be re-read live instead.
Step 3 — Write the reconciliation	Produce the single reconciled entry. No appended rival version; what remains true survives.
Step 4 — State the record update	Name exactly what changed in the record, and what was deliberately not recorded.

Identity lives in the record — not in a session

> **“Memory preserves identity; live systems preserve truth.”**

Principles

The standing rules survive every session because they are held in the record, not in a conversation.

Records

Decisions, commitments, and outcome history — the estate’s past, kept usable.

Role contracts

Who owns what, and what done means — identity as structure, not recollection.

Acceptance standards

The tests work must pass. A system is what it refuses to accept as finished.

Identity does not live in a single model session. Sessions end; the curated record is what makes the next session the same system.

Full checkpoint — the whole session, cold

1	Three classes, contents, and read policies	2	The storage map — four addresses, owners, rules
3	The security limitation, verbatim	4	The recall cycle — four moves, in order
5	The reconciliation rule as a write protocol	6	The continuity rule and the two authorities
7	Control, not convenience — the three failures	8	The publication boundary and its three nevers

The whole session, cold. This is the standard the closing self-check is marked against.

Two acts, one line each

Act I in one sentence

Every kind of state has one home and one read policy: durable memory for how we work, working state owned by its goal or service, live source re-read at decision time.

Act II in one sentence

Memory means curated, reconciled, bounded state: one current record, two authorities that never overlap, and a boundary that keeps the private life inside the estate.

What memory guarantees — three paired promises

Identity without stale state

The estate stays itself across sessions — because identity is carried by the curated record, not by yesterday's values.

Continuity without impersonation

Prior work resumes seamlessly — while remembered state never presents itself as the current read.

Control without convenience

The memory / live distinction binds under pressure — a control that holds exactly when bypassing it would be easiest.

Closing position — memory as identity, kept honest

> **“Memory preserves identity; live systems preserve truth.”**

Curated, reconciled, bounded: memory is the estate's identity, kept honest. It remembers how the estate works, defers to live systems for what is true, corrects itself by reconciliation rather than accumulation, and keeps the private life it holds permanently inside the estate.

Self-check — memory, defended cold

1	The three persistence classes and their read policies
2	The storage map — four addresses and their owners
3	The security limitation, verbatim
4	The recall cycle — four moves, in order
5	The reconciliation rule, applied to a live correction
6	The continuity rule and the two authorities
7	The distinction as a control, not a convenience
8	The publication boundary and its three nevers

Defend each line cold. Anything that cannot be defended today is next session's opening question.